



# Crash Course in Maths for Cognitive Science

## Session 5 | Probability & Statistics for Modelling

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## Session 5 | Probability & Statistics for Modelling

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## What today is about

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A refresher of the probability and statistics from your bachelor, brought back into working order and pushed one measured step further into the tools of inference.

- *Random variables*, the core *distributions*, *expectation* and *variance*.
- *Covariance*, joint distributions, and the *multivariate Gaussian*.
- The *Normal*, the *law of large numbers*, and the *central limit theorem*.
- *Estimation*, *maximum likelihood*, and *confidence intervals*.
- *Hypothesis testing* (z, t, two-sample), *Bayesian inference*, and a glimpse of modelling.

## Definition

A **random variable**  $X$  assigns a number to each outcome of a random experiment. It is *discrete* if it takes countably many values (a die, a count of successes) and *continuous* if it takes values in a range (a height, a reaction time).

## Intuition

The random variable is the “measurement”; the *distribution* is the rule that says how probable each value is. Everything else today is a property of that distribution.

## How a distribution is described

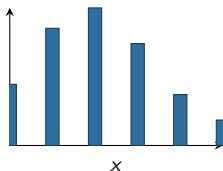
### Useful fact

- **Discrete:** a *probability mass function*  $p(x) = \mathbb{P}(X = x)$ , with  $\sum_x p(x) = 1$ .
- **Continuous:** a *probability density*  $f(x) \geq 0$  with  $\int f(x) dx = 1$ ; probabilities are *areas*,  $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx$ .
- **Either:** the *cumulative distribution*  $F(x) = \mathbb{P}(X \leq x)$ , increasing from 0 to 1.

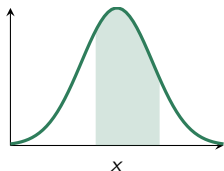
### Common pitfall

For a continuous variable,  $\mathbb{P}(X = x) = 0$  for any single point; only intervals carry probability. A density  $f(x)$  can even exceed 1; it is an area, not a probability.

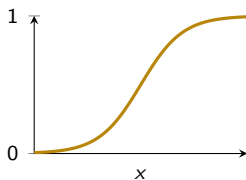
## Three pictures: mass, density, cumulative



pmf (discrete)



pdf (area = probability)



cdf (climbs  $0 \rightarrow 1$ )

The cdf is the running total of the mass or area to the left, so it never decreases.

## The distributions worth keeping at your fingertips

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| Distribution              | Models                     | Mean / Variance                             |
|---------------------------|----------------------------|---------------------------------------------|
| Bernoulli( $p$ )          | one yes/no trial           | $p$ / $p(1 - p)$                            |
| Binomial( $n, p$ )        | # successes in $n$ trials  | $np$ / $np(1 - p)$                          |
| Poisson( $\lambda$ )      | # rare events in a window  | $\lambda$ / $\lambda$                       |
| Uniform( $a, b$ )         | "no value preferred"       | $\frac{a+b}{2}$ / $\frac{(b-a)^2}{12}$      |
| Normal( $\mu, \sigma^2$ ) | sums of many small effects | $\mu$ / $\sigma^2$                          |
| Exponential( $\lambda$ )  | waiting time to an event   | $\frac{1}{\lambda}$ / $\frac{1}{\lambda^2}$ |

The first three are discrete, the last three continuous. If you recognise which one a situation calls for, most problems become lookups.

## The Binomial: try it

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### Example

A participant does 5 independent trials, each correct with probability  $p = 0.4$ . Let  $X$  be the number correct. What is  $\mathbb{P}(X = 2)$ , and what are the mean and variance of  $X$ ?

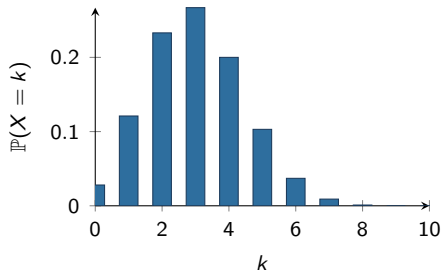
*Try it yourself first — the solution is on the next slide.*

## The Binomial: solution

$X \sim \text{Binomial}(5, 0.4)$ , so

$$\mathbb{P}(X = 2) = \binom{5}{2} (0.4)^2 (0.6)^3 \approx 0.35.$$

Mean  $np = 2$ , variance  $np(1 - p) = 1.2$ . The mass function is a bump peaking near the mean, as on the right (here  $n = 10$ ,  $p = 0.3$ ).



Binomial(10, 0.3) mass function.

# From Binomial to Poisson, and waiting times

## Key idea

When  $n$  is large and  $p$  small, with  $\lambda = np$  moderate, the Binomial converges to the **Poisson**( $\lambda$ ), the law of counts of rare events in a fixed window (calls per minute, spikes per second).

## Intuition

If events arrive as a Poisson process at rate  $\lambda$ , the *waiting time* between them is **Exponential** with mean  $1/\lambda$ . For  $\lambda = 3$  per minute:  $\mathbb{P}(\text{no event in a minute}) = e^{-3} \approx 0.05$ , and the mean wait is 20 seconds. Counts are Poisson, gaps are Exponential, two sides of one coin.

### Practice — try it

1. You flip a fair coin 10 times. Which distribution describes the number of heads, and what are its mean and variance?
2. A call centre receives on average 3 calls per minute. Which distribution fits the count in one minute, and which fits the gap between calls?
3. Is “reaction time in milliseconds” a discrete or a continuous variable?

## Expectation: the long-run average

### Definition

The **expectation** (mean) is the probability-weighted average value:

$$\mathbb{E}[X] = \sum_x x p(x) \quad (\text{discrete}), \quad \mathbb{E}[X] = \int x f(x) dx \quad (\text{continuous}).$$

### Intuition

$\mathbb{E}[X]$  is where the distribution “balances”, the average you would approach if you repeated the experiment many times. It need not be a value  $X$  can actually take (the mean of a die is 3.5).

## Variance: the spread

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### Definition

The **variance** is the average squared distance from the mean, and the **standard deviation** is its square root (back in the original units):

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

### Intuition

Variance measures how widely values scatter around the mean. Two distributions can share a mean yet look completely different because their variances differ.

### Useful fact

- **Linearity of expectation** (always, even for dependent variables):  
 $\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$
- **Scaling/shift of variance:**  $\text{Var}(aX + b) = a^2 \text{Var}(X)$ ; a shift does nothing, a scale squares.
- **Sums of *independent* variables:**  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$

## Expectation and variance: try it

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### Example

Let  $X$  be the result of one roll of a fair six-sided die. Find  $\mathbb{E}[X]$  and  $\text{Var}(X)$ .

*Try it yourself first — the solution is on the next slide.*

## Expectation and variance: solution

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Each face has probability  $\frac{1}{6}$ , so

$$\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5.$$

For the variance,  $\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6}$ , hence

$$\text{Var}(X) = \mathbb{E}[X^2] - \mu^2 = \frac{91}{6} - 3.5^2 = \frac{35}{12} \approx 2.92.$$

So  $\text{SD}(X) \approx 1.71$ , the typical distance of a roll from the average 3.5.

### Definition

The **covariance** measures how two variables move together, and the **correlation** rescales it to  $[-1, 1]$ :

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)], \quad \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}.$$

### Intuition

Positive covariance: when one is above its mean, so is the other. Zero: no *linear* link. Correlation strips out the units and scale, so  $+1$  is a perfect increasing line,  $-1$  a perfect decreasing one.

### Example

Four paired observations:  $(1, 1)$ ,  $(2, 3)$ ,  $(3, 2)$ ,  $(4, 4)$ . Find the correlation of  $X$  and  $Y$ , and then the variance of  $X + Y$ . *Useful:*  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$ .

*Try it yourself first — the solution is on the next slide.*

## Covariance: solution

Both means are 2.5. Using  $n - 1 = 3$ :  $\text{Var}(X) = \text{Var}(Y) = \frac{5}{3}$  and  $\text{Cov}(X, Y) = \frac{4}{3}$ , so

$$\text{Corr}(X, Y) = \frac{4/3}{5/3} = 0.8.$$

And  $\text{Var}(X + Y) = \frac{5}{3} + \frac{5}{3} + 2 \cdot \frac{4}{3} = 6$ .

### Intuition

Because the two rise together (positive covariance), their sum varies *more* than if they were independent (which would give  $\frac{5}{3} + \frac{5}{3} \approx 3.3$ ). Correlation of 0.8 is a strong but imperfect linear link.

### Practice — try it

1. If  $\text{Var}(X) = 4$ , what is  $\text{Var}(3X + 5)$ ?
2.  $X, Y$  independent with variances 2 and 3: what is  $\text{Var}(X - Y)$ ?
3. If  $\text{Corr}(X, Y) = 0$ , must  $X$  and  $Y$  be independent?

## Two variables at once

### Definition

The **joint** distribution  $p(x, y)$  gives the probability of pairs. From it, the **marginal**  $p(x) = \sum_y p(x, y)$  describes one variable alone, and the **conditional**  $p(y | x) = p(x, y)/p(x)$  describes one given the other.

### Intuition

Marginal: ignore the other variable and sum it out. Conditional: fix the other variable and renormalise. The joint contains everything; marginals and conditionals are two ways of reading it.

# Independence, and its weaker cousin

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## Key idea

$X$  and  $Y$  are **independent** when  $p(x, y) = p(x) p(y)$  for all values: knowing one tells you nothing about the other. Independence implies zero covariance.

## Common pitfall

The converse fails in general: *uncorrelated does not mean independent*. Correlation only sees *linear* association, so a strong nonlinear link (for example  $Y = X^2$ ) can have zero correlation. The one clean exception is the Gaussian, where uncorrelated *does* imply independent.

# The covariance matrix

## Definition

For a vector of variables, the **covariance matrix**  $\Sigma$  collects every variance and covariance:

$$\Sigma = \begin{pmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{pmatrix}.$$

It is *symmetric* and *positive-semidefinite* (no direction has negative variance).

## Intuition

The diagonal holds the spreads, the off-diagonal the couplings. This is the exact object PCA diagonalises: its eigenvectors are the principal axes of the data cloud.

# The multivariate Gaussian

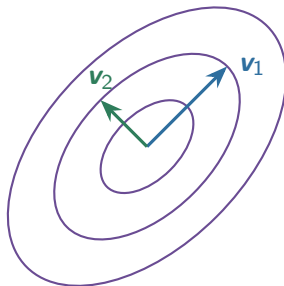
## Key idea

$\mathcal{N}(\mu, \Sigma)$  generalises the bell curve to several variables. It is fixed by a **mean vector**  $\mu$  (the centre) and the **covariance matrix**  $\Sigma$  (the shape). Its equal-probability contours are *ellipses* whose axes are the eigenvectors of  $\Sigma$  and whose half-widths grow like  $\sqrt{\lambda}$  of its eigenvalues.

Now wearing a probabilistic hat: the covariance matrix is where statistics and linear algebra become one subject.

# The covariance ellipse

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Equal-probability contours of a 2-D Gaussian. The ellipse axes are the eigenvectors  $\mathbf{v}_1, \mathbf{v}_2$  of the covariance matrix; their lengths grow with  $\sqrt{\lambda_1}, \sqrt{\lambda_2}$ . Identical to the PCA picture.

## Reading a covariance matrix: try it

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### Example

A 2-D Gaussian has covariance  $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ . In which directions do its contours stretch, and in what ratio?

*Try it yourself first — the solution is on the next slide.*

## Reading a covariance matrix: solution

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The eigenpairs are  $\lambda_1 = 3$  along  $(1, 1)$  and  $\lambda_2 = 1$  along  $(1, -1)$ . So the contour ellipses have their *long* axis along  $(1, 1)$  and short axis along  $(1, -1)$ , with half-widths in ratio

$$\sqrt{\lambda_1} : \sqrt{\lambda_2} = \sqrt{3} : 1 \approx 1.73.$$

### Intuition

Positive covariance tilts the cloud along  $(1, 1)$ : the two variables tend to rise and fall together. Eigenvector gives the tilt, eigenvalue the spread, reading the shape of data straight off  $\Sigma$ .

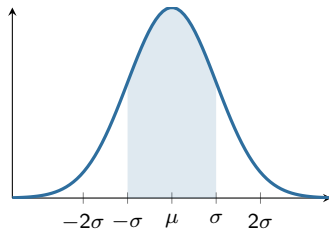
# The Normal distribution

## Definition

$X \sim \mathcal{N}(\mu, \sigma^2)$  has the bell-shaped density

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

$\mu$  sets the centre,  $\sigma$  the width. Its famous rule:  
about 68% / 95% / 99.7% of the mass lies within  
1 / 2 / 3 standard deviations of the mean.



The central 68% band ( $\pm 1\sigma$ ).

## Standardising: the z-score

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### Key idea

Subtract the mean and divide by the standard deviation to get a **standard normal**  $Z \sim \mathcal{N}(0, 1)$ :

$$Z = \frac{X - \mu}{\sigma}.$$

A z-score says how many standard deviations a value sits from the mean, so a single table of  $Z$  answers questions about *any* Normal.

## The z-score: try it

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### Example

IQ scores are modelled as  $\mathcal{N}(100, 15^2)$ . What fraction of people score above 130?

*Try it yourself first — the solution is on the next slide.*

## The z-score: solution

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Standardise the threshold:

$$z = \frac{130 - 100}{15} = 2.$$

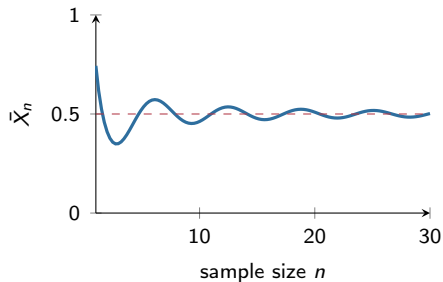
So we need  $\mathbb{P}(Z > 2)$ . From the 68–95–99.7 rule, 95% lies within  $\pm 2\sigma$ , leaving 5% in the two tails, or about 2.5% in each. The exact value is  $\mathbb{P}(Z > 2) = 2.28\%$ , roughly one person in forty.

# The law of large numbers

## Key idea

As the sample size grows, the **sample mean**  $\bar{X}_n$  converges to the true mean  $\mu$ . Averaging many observations cancels the noise.

This is why repeating and averaging works, and why a long-run relative frequency settles down to a probability.



A running average settling to the true mean.

# The central limit theorem

## Key idea

For large  $n$ , the sample mean is approximately **Normal**,

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right),$$

*whatever* the shape of the original distribution. Its standard deviation, the **standard error**, is  $SE = \sigma/\sqrt{n}$ .

## Intuition

The LLN says the average settles; the CLT says how fast and in what shape. This is the engine behind every confidence interval and test that follows.

# The central limit theorem, in pictures

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$n = 1$  (flat)



$n = 2$  (triangular)



large  $n$  (bell)

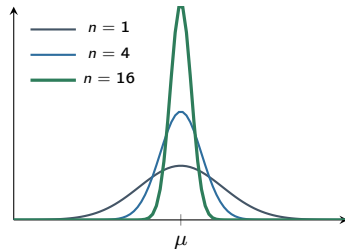
Averaging just a few draws already pulls a flat distribution toward the bell; the more you average, the closer it gets.

# Why the standard error shrinks like $1/\sqrt{n}$

## Intuition

Because  $SE = \sigma/\sqrt{n}$ , quadrupling the sample only halves the error; precision is expensive, but guaranteed. Note  $\sigma$  itself does not change with  $n$ , only the spread of the *mean* does.

The sampling distribution of the mean is (approximately) Normal with a known width, which is exactly what makes intervals and tests possible.



The mean's distribution narrows as  $n$  grows.

### Practice — try it

1. A score is 1.5 standard deviations above the mean. What is its z-score, and is it in the top 10%?
2. Measurements have  $\sigma = 15$ . What is the standard error of the mean of  $n = 25$  of them?
3. You average a very skewed variable over a large sample. What shape does the average have, approximately?

### Definition

An **estimator** is a rule that turns a sample into a guess for an unknown quantity. The two everyday ones are the **sample mean** and **sample variance**:

$$\bar{x} = \frac{1}{n} \sum_i x_i, \quad s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2.$$

### Intuition

An estimate is itself random; a different sample gives a different number. So we care not just about the estimate but about how much it would wobble from sample to sample, its standard error.

### Key idea

An estimator is **unbiased** if it is right *on average* over many samples. The sample mean is unbiased for  $\mu$ . Dividing the squared deviations by  $n$  would *underestimate* the variance, because deviations are taken about the sample mean (which sits inside the data). Dividing by  $n - 1$  corrects exactly for this.

The “ $-1$ ” is the one *degree of freedom* spent estimating the mean, and it comes back as the degrees of freedom of the  $t$ -distribution shortly.

# Maximum likelihood: the fitting principle

## Definition

Given a model  $p(x | \theta)$  and independent data, the **likelihood** and its logarithm are

$$L(\theta) = \prod_i p(x_i | \theta), \quad \ell(\theta) = \sum_i \log p(x_i | \theta).$$

The **maximum-likelihood estimate** is the  $\theta$  that maximises  $\ell$ .

## Intuition

Likelihood flips the usual question: fix the data we saw and ask which  $\theta$  makes it most plausible. The log turns a product into a sum without moving the maximum.

## Maximum likelihood: try it

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### Example

A participant succeeds on  $k$  of  $n$  independent trials, modelled as  $\text{Bernoulli}(p)$ . Find the maximum-likelihood estimate  $\hat{p}$ , and evaluate it for  $k = 8$ ,  $n = 10$ .

*Try it yourself first — the solution is on the next slide.*

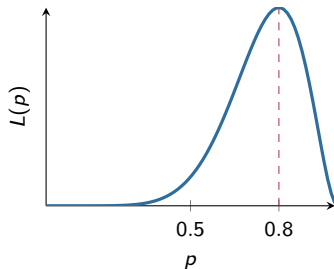
## Maximum likelihood: solution

The log-likelihood is

$\ell(p) = k \ln p + (n - k) \ln(1 - p)$ . Set  $\ell'(p) = 0$ :

$$\frac{k}{p} - \frac{n - k}{1 - p} = 0 \implies \hat{p} = \frac{k}{n} = 0.8.$$

Maximum likelihood recovers the obvious estimate, the observed success rate, but by a recipe that keeps working when the answer is not obvious.



$k = 8, n = 10$ : peak at  $\hat{p} = 0.8$ .

## Maximum likelihood recovers familiar estimates

### Key idea

For Gaussian data, the MLE of the mean is the *sample mean* and the MLE of the variance is the *mean squared deviation*. And fitting a line by **least squares** is exactly maximum likelihood under Gaussian noise.

### Example

For  $\{2, 4, 4, 4, 5, 5, 7, 9\}$  the Gaussian MLEs are  $\hat{\mu} = 5$  and  $\hat{\sigma}^2 = \frac{32}{8} = 4$  ( $\hat{\sigma} = 2$ ). The everyday summaries are maximum likelihood in disguise.

## Confidence intervals (known $\sigma$ )

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### Definition

A 95% **confidence interval** is a range built from the data so that, over repeated samples, 95% of such intervals contain the true parameter. For a mean with known  $\sigma$ ,

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} \quad (1.96 \text{ from the Normal}).$$

### Common pitfall

The interval is random, not the parameter. “95% confidence” describes the *procedure* (95% of intervals catch the truth), not a probability that this one particular interval does.

## A confidence interval: try it

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### Example

From  $n = 25$  measurements with known  $\sigma = 15$  you obtain  $\bar{x} = 100$ . Give a 95% confidence interval for the true mean  $\mu$ .

*Try it yourself first — the solution is on the next slide.*

## A confidence interval: solution

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The standard error is  $SE = \sigma/\sqrt{n} = 15/5 = 3$ , so

$$100 \pm 1.96 \times 3 = 100 \pm 5.88 = [94.1, 105.9].$$

### Intuition

Wider  $\sigma$  or smaller  $n$  widens the interval; more data shrinks it like  $1/\sqrt{n}$ . But this used a *known*  $\sigma$ , which we rarely have, that is what the  $t$ -distribution fixes next.

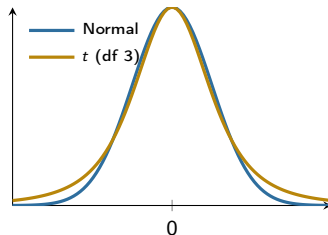
## When $\sigma$ is unknown: the $t$ -distribution

### Key idea

With  $\sigma$  unknown we estimate it by  $s$ , which adds uncertainty. The standardised mean then follows a  **$t$ -distribution** with  $n - 1$  degrees of freedom, like the Normal but with *heavier tails*. The CI becomes

$$\bar{x} \pm t_{0.025, n-1} \frac{s}{\sqrt{n}}.$$

As  $n$  grows,  $t$  approaches the Normal (and  $t_{0.025} \rightarrow 1.96$ ).



The  $t$  has heavier tails.

## A $t$ -interval: try it and check

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### Example

A sample of  $n = 10$  reaction times has  $\bar{x} = 52$  and sample SD  $s = 6$ . Give a 95% confidence interval for the mean. Use  $t_{0.025, 9} = 2.262$ .

The standard error is  $s/\sqrt{n} = 6/\sqrt{10} \approx 1.90$ , so

$$52 \pm 2.262 \times 1.90 = 52 \pm 4.29 = [47.7, 56.3].$$

Slightly wider than a  $z$ -interval would be (1.96 vs 2.262), the price of not knowing  $\sigma$ .

# The logic of a test

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## Key idea

State a **null hypothesis**  $H_0$  (“nothing is going on”) and an **alternative**  $H_1$ . Assume  $H_0$ , then ask: how surprising is the data we actually saw? If it is too surprising under  $H_0$ , we reject  $H_0$ .

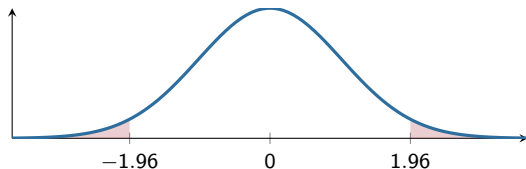
## Intuition

A hypothesis test is a proof by contradiction with a probability attached. We never prove  $H_0$  true; we only find the data compatible with it, or surprising enough to doubt it.

## Test statistic, $p$ -value, significance

### Definition

A **test statistic** measures how far the data sit from what  $H_0$  predicts, in standard-error units. The  **$p$ -value** is the probability, *if  $H_0$  were true*, of a statistic at least this extreme. Reject  $H_0$  when  $p < \alpha$ , the pre-chosen **significance level** (often 0.05).



Two-tailed test at  $\alpha = 0.05$ : reject if  $|z| > 1.96$ .

## A one-sample z-test: try it

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### Example

A standard task has mean 100. A group of  $n = 25$  people scores  $\bar{x} = 106$ ; the population standard deviation is  $\sigma = 15$ . Test  $H_0 : \mu = 100$  against  $H_1 : \mu \neq 100$  at  $\alpha = 0.05$ .

*Try it yourself first — the solution is on the next slide.*

## A one-sample z-test: solution

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The standard error is  $SE = 15/\sqrt{25} = 3$ , so

$$z = \frac{\bar{x} - \mu_0}{SE} = \frac{106 - 100}{3} = 2.0.$$

Since  $|z| = 2.0 > 1.96$ , we reject  $H_0$  at  $\alpha = 0.05$  (two-tailed  $p \approx 0.046$ ).

### Intuition

Two standard errors above the null value is surprising enough, if nothing were going on, to doubt  $H_0$ . But the group could still differ by a trivial amount: *statistical* significance is not *practical* importance.

## The $t$ -test: unknown $\sigma$

### Example

A more realistic version:  $n = 16$ ,  $\bar{x} = 105$ , and the *sample* SD is  $s = 8$  (we do not know  $\sigma$ ). Test  $H_0 : \mu = 100$  at  $\alpha = 0.05$ .

The statistic uses  $s$  and the  $t$ -distribution with  $n - 1 = 15$  df:

$$t = \frac{105 - 100}{8/\sqrt{16}} = \frac{5}{2} = 2.5, \quad t_{0.025, 15} = 2.131.$$

Since  $2.5 > 2.131$ , reject  $H_0$  ( $p \approx 0.025$ ). With a small sample and an estimated  $\sigma$ , the  $t$ -test is the honest choice.

## Comparing two groups

### Example

Group A ( $n = 30$ , mean 54) versus group B ( $n = 30$ , mean 48), each with SD 8. Is the difference significant?

The standard error of the difference is  $\sqrt{8^2/30 + 8^2/30} \approx 2.07$ , so

$$t = \frac{54 - 48}{2.07} \approx 2.9 \quad (p \approx 0.005).$$

### Intuition

Comparing two means means testing whether their *difference* is far from zero, in units of the standard error of that difference. Variances add, so the two groups each contribute to the noise.

## Two ways to be wrong

### Useful fact

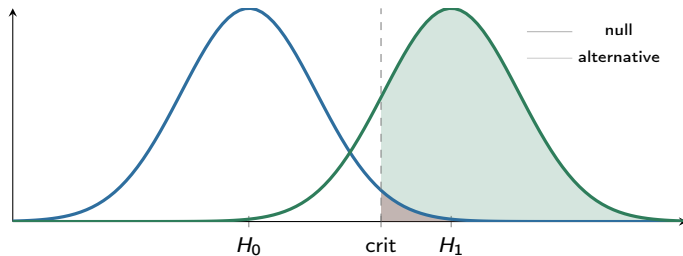
- **Type I ( $\alpha$ ):** reject  $H_0$  when it is true, a false alarm.
- **Type II ( $\beta$ ):** fail to reject  $H_0$  when  $H_1$  is true, a miss.
- **Power** =  $1 - \beta$ : the chance of catching a real effect; it grows with sample size and effect size.

|              | $H_0$ true | $H_1$ true |
|--------------|------------|------------|
| reject $H_0$ | Type I     | ✓ power    |
| keep $H_0$   | ✓          | Type II    |

### Common pitfall

A large  $p$ -value is *not* evidence that  $H_0$  is true, only a failure to reject it.

## Power, in a picture



Red: Type-I probability  $\alpha$  (null past the critical value). Green: power (alternative past the critical value).  
Moving  $H_1$  further right, or shrinking the curves with more data, raises power.

### Practice — try it

1. You double the sample size. By what factor does a confidence interval's width change?
2. When should you use a  $t$ -test rather than a  $z$ -test?
3. A test gives  $p = 0.20$ . Have you shown there is no effect? Name one way to raise power.

## Conditional probability, and reversing it

### Definition

The **conditional probability** of  $A$  given  $B$  is  $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$ . Rearranged,  $\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$ .

### Intuition

Often we know one direction,  $\mathbb{P}(\text{positive test} \mid \text{disease})$ , but want the other,  $\mathbb{P}(\text{disease} \mid \text{positive test})$ . Bayes' theorem flips the conditioning around.

### Key idea

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \mathbb{P}(A)}{\mathbb{P}(B)}, \quad \mathbb{P}(B) = \mathbb{P}(B \mid A) \mathbb{P}(A) + \mathbb{P}(B \mid \bar{A}) \mathbb{P}(\bar{A}).$$

The denominator (the *total probability* of  $B$ ) adds up all the ways  $B$  can happen. The numerator keeps the one path through  $A$ . Their ratio is the updated probability of  $A$  once we have learned  $B$ .

## The base-rate example: try it

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### Example

A disease affects 1% of people. A test detects it in 99% of those who have it (sensitivity), but also gives a false positive in 5% of those who do not. You test positive. What is the probability you actually have the disease?

*Try it yourself first — the solution is on the next slide.*

## The base-rate example: solution

Total probability of a positive test:

$$\mathbb{P}(+) = \underbrace{0.99 \cdot 0.01}_{\text{true } +} + \underbrace{0.05 \cdot 0.99}_{\text{false } +} = 0.0594.$$

Bayes' theorem then gives

$$\mathbb{P}(D \mid +) = \frac{0.99 \cdot 0.01}{0.0594} \approx 0.17.$$

Only about **17%**: the disease is rare, so most positives are false positives.

| per 10,000     | test + | test - |
|----------------|--------|--------|
| disease (100)  | 99     | 1      |
| healthy (9900) | 495    | 9405   |
| total          | 594    | 9406   |

Of 594 positives, only 99 are truly ill:

$$99/594 \approx 17\%.$$

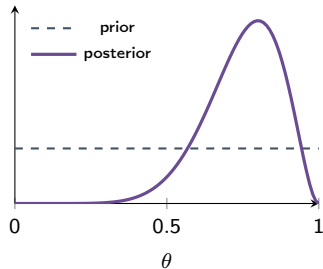
## From events to a parameter

The same rule works when the unknown is a continuous *parameter*  $\theta$ :

$$\underbrace{p(\theta | D)}_{\text{posterior}} \propto \underbrace{p(D | \theta)}_{\text{likelihood}} \times \underbrace{p(\theta)}_{\text{prior}}.$$

### Intuition

Start with a prior belief over  $\theta$ , reweight it by how well each value predicted the data, renormalise. The answer is a whole *distribution*: our belief after the data, uncertainty included.



Data sharpen a belief about  $\theta$ .

## Conjugacy: why Beta meets the Binomial

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Put a  $\text{Beta}(a, b)$  prior on a probability against binomial data of  $h$  hits and  $t$  misses. Multiply prior by likelihood:

$$\theta^{a-1}(1-\theta)^{b-1} \times \theta^h(1-\theta)^t = \theta^{(a+h)-1}(1-\theta)^{(b+t)-1} = \text{Beta}(a+h, b+t).$$

### Key idea

The posterior is *another Beta*: just **add the hits and misses to the prior counts**. Read  $a-1$ ,  $b-1$  as “prior hits and misses”, so  $\text{Beta}(1, 1)$  is a flat prior, “no information yet”.

### Example

A detection probability  $\theta$  has a flat prior  $\text{Beta}(1, 1)$ . You observe 8 detections in 10 trials. Find the posterior, its mean, and its peak (MAP).

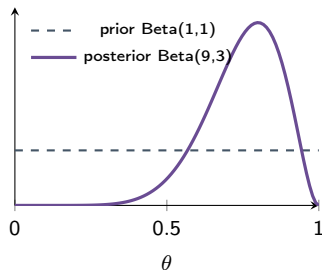
*Try it yourself first — the solution is on the next slide.*

## Bayesian estimation: solution

Add 8 hits and 2 misses to the prior counts:  
Beta(9, 3), with

$$\text{mean} = \frac{9}{12} = 0.75, \quad \text{MAP} = \frac{9-1}{12-2} = 0.80.$$

The peak sits at the MLE (0.8); the mean is pulled a little toward the prior centre. A 95% **credible interval** is about [0.48, 0.94]: we can say directly that  $\theta$  lies there with 95% probability.



Flat prior to peaked posterior.

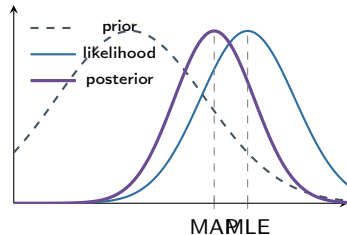
# MLE, MAP, and the credible interval

Three summaries, one posterior:

- **MLE:** the peak of the *likelihood* (data alone).
- **MAP:** the peak of the *posterior* (likelihood  $\times$  prior).
- **credible interval:** a range holding, say, 95% of the posterior.

## Intuition

MAP is the MLE nudged by the prior (they coincide under a flat prior). Unlike a confidence interval, a credible interval is a genuine probability statement about  $\theta$ .



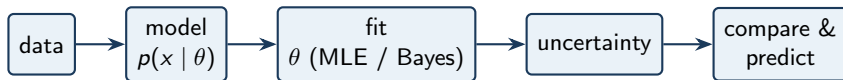
$$\text{Prior} \times \text{likelihood} = \text{posterior}.$$

### Practice — try it

1. In the disease example, if the base rate rose to 10%, would  $\mathbb{P}(D \mid +)$  go up or down?
2. Starting from  $\text{Beta}(1, 1)$ , what is the posterior after 3 hits in 3 trials, and its mean?
3. In one line, how does a credible interval differ from a confidence interval?

## A glimpse ahead: modelling cognition

Everything so far *describes* data. A **cognitive model** goes one step further: it proposes *how* the data were generated, a mechanism with a few **parameters** we estimate from behaviour.



### Intuition

The parameters *are* the psychology: a learning rate, a decision threshold, a perceptual bias. Distributions describe the data, likelihood fits the parameters, and the CLT, intervals, and Bayes quantify the uncertainty. You already own every piece.

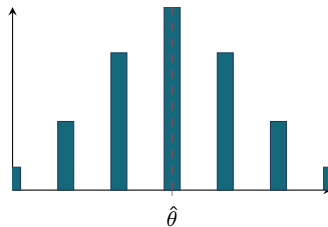
# Uncertainty without formulas: the bootstrap

## Key idea

When no neat standard-error formula exists, **resample the data with replacement** many times, recompute the statistic, and use the spread of those values as its standard error.

## Intuition

The sample stands in for the population, so resampling from it mimics drawing fresh data. It works even for medians or correlations, where no textbook formula exists.



Distribution of a statistic across resamples.

## Session 5: what we covered

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- *Random variables*, the core *distributions*, *expectation*, *variance*, and *covariance*.
- *Joint distributions* and the *multivariate Gaussian*, whose covariance matrix is the PCA/ellipse picture.
- The *Normal*, the *LLN* and *CLT*, with  $SE = \sigma/\sqrt{n}$ .
- *Estimation*, *maximum likelihood*, and *confidence intervals*; the *t*-distribution when  $\sigma$  is unknown.
- *Hypothesis testing* (z, t, two-sample), and *Bayesian inference* with priors, posteriors, MAP, and credible intervals.
- A glimpse of *modelling*, and the *bootstrap* for uncertainty.

These are the working tools you will reach for across data analysis, statistics, and the modelling courses ahead.